PROG2002 – WEB DESIGN AND DEVELOPMENT

A-03 : *ROCK,PAPER,SCISSORS,LIZARD,SPOCK* – WEBSITE DEVELOPED USING CGI AND PHP

OVERVIEW

In this assignment, you will create a simple two-page web site that uses a form to provide the parameters to a web server. You will implement both the CGI and PHP versions of this website…

This is a partner-based assignment and you can have partner if you wish.

OBJECTIVES

This assignment supports the following course objectives:

- To demonstrate the operation of server side technologies (CGI and PHP)
- To demonstrate the ability to create user web forms

ACADEMIC INTEGRITY AND LATE PENALTIES

- Please refer to the SET Policies document regarding Academic Integrity Information
- Please refer to the SET Policies document regarding Late Policy

EVALUATION

- Please refer to the assignment weighting in the *Instructional Plan* for the course as well as the assignment's Rubric in the course shell.

PREPARATION

Review Module-03, Module-04 and Module-05 lesson content as well as each module's code samples – they will help you in this assignment.

REQUIREMENTS

1. In this assignment – you need to create two (2) HTML forms that are identical, except where the CGI implementation uses the form GET method and the PHP implementation uses the form POST method.
2. Each of the implementations start with this form which must:
    a. Have a title banner (header or graphic) showing the title "Rock, Paper, Scissors, Lizard, Spock"
    b. Ask for the user's name and validate input using JavaScript. (i.e. is a mandatory field and not blank – can accept any characters)
    c. In a drop-down list, ask for Big Bang Character to play against
        - The characters are Amy, Sheldon, Penny, Leonard, Bernadette, Howard, Raj and Stuart
        - This is the character that the computer will play as
        - Make sure the first option (default selected) in the dropdown is **blank**
        - You need to validate that the user makes a selection from this dropdown
    d. After getting the user's name and character choice and validating them, you may then display and ask the user (by name) to choose a *game selection* from the list of five choices (rock / paper / scissors / lizard /Spock)
        - Use a set of radio buttons for this control
        - Make sure there are no radio buttons selected (i.e. no default selected)
        - You will need to validate that the user has actually made a selection before submitting
    e. Once validated, the user's name, character choice and the game selection are to be sent to the server as appropriate key value pairs, in each of the two HTML pages, respectively.
    f. Name these HTML files `cgi-rpsls.html` and `php-rpsls.html`.
3. The root folder of your web site on the server is to contain the two HTML pages as well as the PHP server-side page. This is also where your CGI executable needs to be located.
4. On the server-side of programming:
    a. You will need to randomize and choose a random number between 1 and 5 (inclusive) to represent (1) the Rock, (2) the Paper, (3) the Scissors, (4) the Lizard and (5) Spock
    b. This also means you will need to transform the user's *game selection* into a number in the same range
    c. You then need to compare the computer's random choice against the user's game selection to see who has WON the round (or if it is a TIE/DRAW)
        - See Rules of the Game section below
    d. At this point, the server must construct a nicely formatted output message indicating the following information (see Rules of Game section for example outputs):

```
<Character Name> WON or <User Name> WON
  - <User Name> chose <insert user's choice>
```

```
  - <Character Name> chose <insert computer's random choice>
  - insert rule describing why <Character Name> won or <User Name> won
** also be aware that this statement may have to describe why this match is a TIE/DRAW
```

   e. Beside the nicely formatted output of the match results, display an image (graphic) indicating that the user WON, LOST or TIED – the image needs to be displayed <u>beside</u> the output message

5. Create a sub-folder (called `theGame`) and have it exist in your site (create it in the root of your web site).
   a. This directory is to contain the images of the win, lose and tie graphics that you will use in your game play. You can create these images yourself or source them out on the internet.
   b. Each server technology composes the response, with:
      - The neatly formatted output (using a bullet list)
      - The WIN/LOSE/TIE picture of how this match went being displayed (no larger than 400 pixels x 400 pixels)
      - Remember the nicely formatted output is to be placed **<u>beside</u>** the picture
   c. Also remember when referencing your image file to <u>avoid using **absolute file addressing within your website**</u>
6. For the CGI solution, the server file must be called **rpsls.exe** and must be created using the C programming language. For the PHP solution, the server file must be called **rpsls.php**
7. You are not required to make the pages overly fancy, but a reasonable amount of effort should be made to align the content well and also use CSS techniques (inline, internal or external styles can be used) to increase the user experience.
8. Both the CGI and PHP implementations of this website should have **<u>exactly</u>** the same *look and feel*. Meaning – the styling, layout, formatting, etc. of both implementations needs to be the same.
9. Good programming practices must be exhibited as per SET standards. Make sure to comment all source files appropriately
10. Ensure that your *Rock-Paper-Scissors-Lizard-Spock* application runs properly and consistently within the Microsoft Edge and Chrome browsers

FILE NAMING REQUIREMENTS

As mentioned above in the Requirements, the solution for this assignment must be developed and delivered using specific filenames in order to make marking and testing straightforward. The filenames are:

- `cgi-rpsls.html` and `php-rpsls.html`
- `rpsls.exe` and `rpsls.php`

SUBMISSION REQUIREMENTS

When submitting your solution to this assignment, hand-in a single ZIP'd file containing:

1. The two HTML *Rock-Paper-Scissors-Lizard-Spock* launching pages (HTML files)
2. The cleaned Visual Studio solution for the CGI executable file (no need to submit the executable)
    a. Submit the **entire Visual Studio solution** – i.e. ZIP up the entire directory, but **clean** it first
3. The PHP server-side source file
4. The supporting folder [i.e. `theGame` folder] (correctly named to match your code) with appropriate files (images).
5. Also remember that this solution will be tested using Microsoft Edge as well as Chrome
6. Please ZIP up all of these files into a single ZIP file and submit to the appropriate eConestoga Dropbox by the deadline
    a. Please give your ZIP submission the filename *lastName-firstInitial.zip* (e.g. if you are Sally Jones – then your ZIP should be named `jones-s.zip`
    b. If you are working with a partner, then include both your names in the ZIP filename (e.g. if Sally Jones is working with John Smith – then your ZIP should be named `jones-s-smith-j.zip`

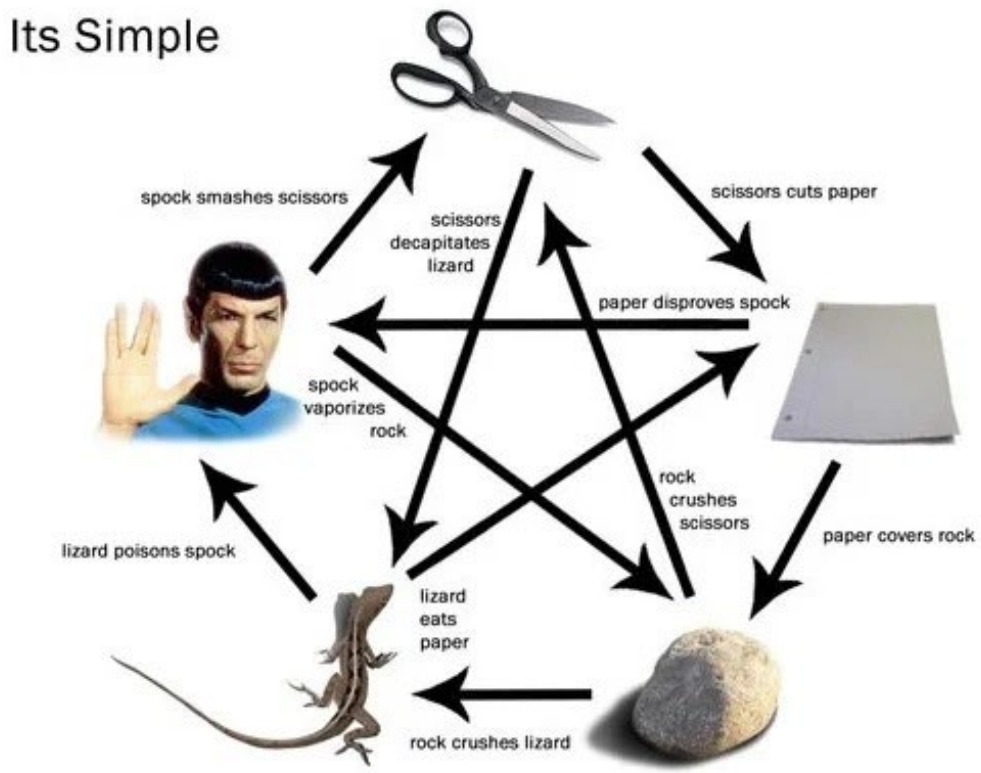
    NOTE: If working with a partner, only one partner need submit the solution

RULES OF THE GAME (AND SAMPLE OUTPUT)

Here are a couple of links to help describe the rules of this game to you – one from Big Bang Theory Famdon and the other is a video explaining the rules/concept:

- Rock, Paper, Scissors, Lizard, Spock Explained
- Video of the Rules

Not sure if it will help – but this image also (pictorially) displays the rules and explains them…



In terms of logic/rules and explanation of the game … here's a table that helps explain…

Logic / Rule	Reason
Scissors WIN over Paper	Scissors cut Paper
Paper WINS over Rock	Paper cover Rock
Rock WINS over Lizard	Rock crushes Lizard
Lizard WINS over Spock	Lizard poisons Spock
Spock WINS over Scissors	Spock smashes Scissors
Scissors WIN over Lizard	Scissors decapitate Lizard
Lizard WINS over Paper	Lizard eats Paper
Paper WINS over Spock	Paper disproves Spock
Spock WINS over Rock	Spock vaporizes Rock
Rock WINS over Scissors	Rock crushes Scissors

** if both the user and the computer make the same choice, then the match is a TIE/DRAW and nobody WINS

Here are some sample matches and sample outputs:

1. Let's say a user named Sally chose to play against Sheldon. Sally chose to use the Rock. And the computer (Sheldon) randomly chose Paper… Here is the output that would result (without the WIN/LOSE/TIE image being shown)

   Sheldon **WON**
   - Sally chose the Rock
   - Sheldon chose Paper
   - According to the rules: *Paper covers Rock*

2. Let's say a user named Bill chose to play against Howard. Bill chose to use the Lizard. And the computer (Howard) randomly chose Spock… Here is the output that would result (without the WIN/LOSE/TIE image being shown)

   Bill **WON**
   - Bill chose the Lizard
   - Howard chose Spock
   - According to the rules: *Lizard Poisons Spock*

3. Let's say a user named Maurice chose to play against Penny. Maurice chose to use the Scissors. And the computer (Penny) randomly chose Scissors… Here is the output that would result (without the WIN/LOSE/TIE image being shown)

   Nobody **WON** – it was a **TIE**
   - Maurice chose the Scissors
   - Penny chose the Scissors
   - According to the rules: *when both players make the same choice, it is a TIE*